One Stack, Many Rankings: Evaluating Quantized Reasoning Checkpoints Beyond Accuracy

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Abstract

Public quantized reasoning checkpoints do not receive a single deployment ranking: under one pinned serving stack, rank order depends on the checkpoint, task, evaluation target, estimand, and serving condition. We evaluate public DeepSeek-R1-Distill checkpoints on an NVIDIA A100-80GB with vLLM 0.7.0 [13] eager execution, varying only the public weight checkpoint among BF16, FP8 executed as Marlin W8A16 [6] rather than native W8A8, AWQ-4, and GPTQ-4 for DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B. The evaluation comprises 88 checkpoint×benchmark×seed runs and 56,408 completions on MATH-500 [10,16] (5 seeds), GSM8K [3] (3 seeds), and GPQA-Diamond [21] (3 seeds).

Observed under this pinned stack, the tested checkpoints change rank across evaluation targets. On MATH-500, FP8–BF16 pass@1 differences are +0.40 and +0.28 percentage points (pp); problem-clustered 95% intervals include zero, and a ±1 pp equivalence test is not passed. The tested community Qwen AWQ-4 artifact showed 5.56 pp lower pass@1 on GPQA-Diamond. The Qwen AWQ GPQA result is significant within the primary Holm-6 family, but not under the Holm-18 joint sensitivity analysis. The tested Qwen 4-bit checkpoints showed 6.3–6.9% higher mean MATH-500 completion length, including a detectable association among jointly correct Qwen AWQ-4/GPTQ-4 pairs; much larger conditional differences occur in mismatch cases, where failure traces are themselves long. Strict five-sample unique-mode agreement yields low observed selective risk, trading off coverage and requiring five generations. The historical token proxy, sequential GPU-seconds (Condition A), and batched GPU-seconds (Condition B) produce different rankings among the tested Qwen cells. The deployment ranking of a quantized reasoning checkpoint is not a property of bit-width alone.

Keywords: Reasoning language models, public quantization checkpoints, pinned serving stack, estimand disagreement, Cost-of-Pass.

1 Introduction

Reasoning models such as DeepSeek-R1 [8] emit long traces before a final answer. Post-training quantization (FP8, AWQ-4 [17], GPTQ-4 [5]) is therefore a common serving choice. Isolated pass@1 numbers hide whether compression also changes trace length [15,19,20], termination, or the cost of a correct answer [4,12]. Serving-stack details—engine version, decoding flags, CUDA graphs—can materially move reasoning scores [11]. This paper *pins* one stack rather than shifting it: we do not run a factorial vLLM 0.7.0 vs. 0.8.5 experiment. The title names multiple rankings under one stack; "beyond accuracy" is a measurement checklist (length, agreement, GPU-seconds), not a claim of a new evaluation method.

The deployment ranking of a quantized reasoning checkpoint is not a property of bit-width alone.

We ask four questions under that pin:

- **RQ1.** Do the evaluated quantized checkpoints differ in *pass@1* from matched BF16, with problem-clustered uncertainty?
- **RQ2.** How do completion length and correctness-conditioned length differ across the evaluated checkpoints, and what do identical-word loops and near-cap completions reveal about the long-tail behavior?
- **RQ3.** What can observable multi-sample agreement say about selective abstention without gold labels at serve time?
- **RQ4.** Do checkpoint rankings agree across the historical token proxy, sequential Condition A, and batched Condition B aggregate serving-cost proxies?

Contributions. (C1) **Pinned evaluation protocol.** NVIDIA A100-80GB; vLLM 0.7.0 eager; public DeepSeek-R1-Distill checkpoints; BF16; FP8 executed as Marlin W8A16; AWQ-4; GPTQ-4; problem-clustered bootstrap; TOST ± 1 pp; and an explicit pass@1 versus maj@5 distinction. (C2) **Empirical characterization of ranking instability across deployment estimands.** Under this pinned stack, the same public checkpoints receive different rankings depending on accuracy, maj@5, correctness-conditioned length, the historical token proxy, sequential GPU-seconds, and batched GPU-seconds. (C3) **Checkpoint-not-method result.** The tested community AWQ artifact showed degradation under this evaluation. That finding applies to the tested artifact; it does not establish a universal property of AWQ. The released grid and analysis artifacts support reproducibility of these contributions.

2 Related Work

GPTQ [5] and AWQ [17] were originally validated mainly on short-form benchmarks. SmoothQuant [23] addresses activation quantization, which we do not run. Liu et al. (QRM; COLM 2025) [19] evaluate DeepSeek-R1 distillations across weight, activation, and KV quantization. QRM reports that *minor-drop* W8/W4A16 settings do not systematically lengthen outputs, whereas more aggressive low-bit settings can. Numerically, QRM reports Qwen-7B MATH BF16 94.60 / AWQ 91.00 and GPQA AWQ 50.00; this pinned grid is 93.12 / 44.78 for the tested Qwen AWQ-4 checkpoint. We therefore interpret the cross-study discrepancy as evidence of checkpoint- and stack-sensitive measurement rather than as a superiority claim. The FP8 and GPTQ-4 cells are public RedHatAI compressed-tensors artifacts released for vLLM serving; we evaluate those checkpoints on A100 Marlin W8A16 and do not treat the measurements as Red Hat production-deployment claims. Lian et al. [15] isolate a BF16-correct length estimand: low-bit PTQ can inflate chain-of-thought even when the uncompressed run is right. Lotfi et al. [20] attribute much of the extra length to overthinking—abandoned intermediate answers rather than missing competence. Li et al. [14] localize 4-bit math failures to early process errors. Alimaskina et al. [1] study loops and delayed commitment under extreme low-bit inference. Kurtic et al. [12] provide a broad accuracy–performance quantization study in vLLM, including Llama-3.1 and DeepSeek-R1-Distill reasoning evaluations and synchronous versus continuous-batching serving regimes. Helcig et al. [9] formalize statistically lossless quantization—task-level versus distribution-level fidelity—and examine reasoning/token and serving behavior under a proposed quantizer. Our object of study is different: we do not propose a quantizer or a fidelity criterion, but ask whether deployment conclusions remain stable across evaluation targets for the same public checkpoints on one frozen A100/vLLM 0.7.0 stack.

Table 1: Neighboring measurement studies. "Engine pinned" means one named serving stack is frozen while checkpoints vary. Length estimands and serving columns are what each paper reports as primary evidence, not an exhaustive inventory.

Study	**Models**	**Bits**	**Engine pinned**	**Seeds**	**Length estimand**	**Correctness split**	**Serving / cost**	**Statistics**
This paper	2 R1 distill (7B/8B)	BF16, FP8, AWQ-4, GPTQ-4	vLLM 0.7.0 eager, A100	5 MATH; 3 GSM/GPQA	RoM, paired Δ, BF16-correct Δ	2×2 + BF16 failure control	token proxy; seq. & batched GPU-s; $\widetilde{C}^{\text{hyb}}_{\text{pass}}$	clustered bootstrap, Holm, TOST
QRM [19]	R1 distill family	W/A/KV incl. W4A16	QRM harness (multi-config)	typically 1	mean tokens	not 2×2	limited	per-cell means
Lian et al. [15]	reasoning PTQ	low-bit PTQ	paper stack	paper-specific	BF16-correct Δ	BF16-correct vs rest	token cost	paired length
Lotfi et al. [20]	1.5B–32B reasoning	aggressive PTQ	paper stack	paper-specific	CoT length; overthinking	abandoned intermediates	logit-penalty Pareto	error-type counts
Kurtic et al. [12]	Llama-3.1 + R1-distill	FP8/INT8/INT4	vLLM, several GPUs	large evals	not CoT-primary	N/A	sync vs continuous batching	accuracy + tok/s
Helcig et al. [9]	Llama/Qwen (SLQ)	3.3–6 bpp	vLLM kernels	3 (GSM)	thinking tokens	TL vs DL fidelity	vLLM throughput	EAR, Wilcoxon
Erol et al. [4]	general LMs	N/A	N/A	N/A	N/A	N/A	C_{pass} definition	economic
Hochlehnert et al. [11]	reasoning LMs	N/A	deliberately *un*pinned	many	N/A	N/A	N/A	reproducibility

Hochlehnert et al. [11] show that seeds, prompts, decoding, and software move reasoning scores; we freeze those factors. Zhong et al. [24] study uncertainty calibration of quantized LLMs using model probabilities—a quantity we do not log. Ayadi et al. [2] study reliability of quantized LLMs via uncertainty, calibration, and input-perturbation robustness; that suite is complementary to, and distinct from, our gold-free unique-mode abstention on five MATH-500 seeds. Self-consistency [22] and G-Pass@k [18] are gold-using repeated-sampling neighbors; our modal rule is a gold-free unique-mode abstention on five MATH seeds, not a new estimator. Selective classification [7] supplies the risk–coverage language. Erol et al. [4] define Cost-of-Pass as expected monetary cost per correct solution. We report a historical shared-65 tok/s token-implied aggregate cost proxy and an aggregate hybrid Cost-of-Pass proxy whose numerator is confirmation GPU cost per query and whose denominator is campaign MATH-500 pass@1; neither is Erol et al.'s per-problem estimator.

Table 1 places this measurement against those studies. The distinctive result here is estimand disagreement under a pinned stack, not a new compressor. The unit of comparison in this study is the public checkpoint $\times$ pinned stack $\times$ evaluation target, rather than a quantization method in isolation.

3 Experimental Methodology

3.1 Models and precision formats

We evaluate DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B [8]. Formats: uncompressed **BF16**; **FP8** checkpoints served on A100 via vLLM's Marlin **W8A16** weight-only fallback (not native W8A8 tensor-core FP8); **AWQ-4** (group size 128, `float16`); **GPTQ-4** (group size 128, Marlin). Hugging Face IDs and revisions are listed in the appendix. AWQ-4 checkpoints are third-party community uploads; accuracy losses on the tested Llama AWQ-4 checkpoint are properties of that artifact, not a proof about AWQ as a method.

3.2 Benchmarks and protocol

MATH-500 [10, 16] ($n = 500$, seeds 42–46): 20,000 completions. GSM8K [3] ($n =$ 1,319, seeds 42–44) and GPQA-Diamond [21] ($n = 198$, seeds 42–44): breadth. Decoding: zero-shot <think> prefix (appendix), $T = 0.6$, top-$p = 0.95$, repetition penalty 1.0 (deliberately not used to hide loops), max new tokens 32,768. Total: 56,408 completions in 88 checkpoint$\times$benchmark$\times$seed runs. Throughout, a *run* denotes one checkpoint$\times$benchmark$\times$seed evaluation; a *cell* denotes one checkpoint$\times$benchmark aggregated over its evaluated seeds; a *format contrast* compares a quantized checkpoint with its matched BF16 checkpoint within the same model family and benchmark. MATH and GSM8K are widely used; a paired within-item design reduces but does not eliminate contamination interaction with format.

3.3 Hardware and serving stack

PARAM Rudra HPC (IIT Patna / NSM), NVIDIA A100-PCIE-80GB. Publication runs use the official QRM `inference.py` path [19] with `gpu_memory_utilization=0.75`. Engine pin:

vLLM==0.7.0, eager execution. Files under configs/models/ and configs/legacy_models/ are *not* the frozen publication launcher; they differ in max_model_len and KV-cache defaults and are not the effective campaign stack. Effective launch settings are recorded in results/reports/runtime_manifest.json.

3.4 Measured serving

After the 56k accuracy campaign we timed the same eight checkpoints on A100-80GB under vLLM 0.7.0 eager. Primary numbers are the *controlled confirmation* (protocol: docs/MEASURED_SERVING_CONFIRMATION_PROTOCOL.md; raw JSON under results/measured_serving_confirmation/raw/). Condition B is the primary deployment-style measured serving condition; Condition A provides a sequential comparator, and the shared 65 tok/s token calculation is retained only as a historical sensitivity proxy. Condition A uses a balanced 20-prompt MATH-500 subset (4 per difficulty level), sequential single-request decode. Condition B uses a balanced 100-prompt subset (20 per level) in one llm.generate call with max_num_seqs=8 pinned. Conditions A and B differ jointly in evaluated prompt subset and serving regime (20 sequential prompts versus 100 continuously batched prompts). Comparisons between them therefore measure sensitivity to the complete serving condition; they do not isolate batching or concurrency as the causal source of a ranking change. Secondary microbenchmark: 10 prompts forced to exactly 512 tokens (ignore_eos). Three warmup queries are discarded. Sampling matches the campaign ($T = 0.6$, top-$p = 0.95$) with seed 20260816. Repetitions share that seed. Aggregate output-token counts are identical across repetitions; therefore the observed throughput spread reflects runtime variation conditional on the same aggregate output-token count. The root cause of the observed slow/mid/fast timing regimes was not identified. The runner starts at $R = 3$ and expands to $R = 5$ when the *population* CV of the first three tok/s values exceeds 3%; this rule allocates two additional technical timing repetitions to cells with unusually high initial wall-clock variation and is a measurement-budget rule rather than an inferential stopping rule for model quality. Reported $\pm$ is *sample* SD. Qwen-7B jobs ran on ragpu003 and Llama-8B jobs on ragpu004 (one visible A100 each; PyTorch 2.5.1+cu124). Cross-architecture absolute tok/s is therefore cautious. An earlier unconstrained timing (docs/MEASURED_SERVING_PROTOCOL.md; mixed nodes; max_num_seqs not pinned) is not averaged with these numbers and is not used in the main tables.

Output tok/s = $(\sum \text{output tokens})/\Delta t_{\text{steady}}$. GPU-sec/query = $\Delta t_{\text{steady}}/N$ on one GPU. Peak VRAM is torch.cuda.max_memory_allocated after the 0.75 utilization preallocation ($\sim 60\,\text{GB}$ reserved of $80\,\text{GB}$). Measured

$$\widetilde{C}^{\text{hyb}}_{\text{pass}} = \frac{(\text{GPU-sec/query}) \times (\$1.50/3600)}{\text{pass@1}}, \tag{1}$$

using the frozen 40-cell MATH-500 means—not accuracy on the 100-item subset. We report an aggregate hybrid Cost-of-Pass proxy, $\widetilde{C}^{\text{hyb}}_{\text{pass}}$. Its numerator is mean confirmation GPU-seconds per query on the serving subset and its denominator is aggregate campaign MATH-500 pass@1. This aggregate ratio is inspired by Cost-of-Pass but is not the per-problem estimator defined by Erol et al. [4]. Serving prompts are not re-scored. Uncertainty intervals are 95% intervals conditional on the observed timing repetitions and a problem-clustered bootstrap of campaign pass@1 ($B = 10{,}000$). The \$1.50/A100-hour rate is a cloud pricing *scenario*, not a billed cluster price.

3.5 Metrics

Pass@1. Mean extractive match across seeds. Primary inference is a problem-clustered bootstrap of the paired difference $\Delta = \overline{\text{acc}}_{\text{quant}} - \overline{\text{acc}}_{\text{BF16}}$. For each problem i we first average correctness over seeds, $\delta_i = \overline{\mathrm{acc}}_{i,\mathrm{quant}} - \overline{\mathrm{acc}}_{i,\mathrm{BF16}}$, so all seeds belonging to a problem are retained in that item mean and treated as fixed rather than independently resampled. We then resample n problems with replacement (B = 10,000; RNG seed 0). The 95% intervals are percentile bootstrap intervals. Accordingly, these intervals quantify variation over benchmark problems conditional on the evaluated decoding-seed set and decoding configuration; decoding seeds are not treated as an additional sampled random effect. With bootstrap replicates Δ_b^*, the two-sided bootstrap tail-area p-value is

$$p_{\mathrm{hi}} = \#\{\Delta_b^* \geq 0\}/B, \quad p_{\mathrm{lo}} = \#\{\Delta_b^* \leq 0\}/B, \quad p = \min\bigl(1,\, 2\min(p_{\mathrm{hi}}, p_{\mathrm{lo}})\bigr). \tag{2}$$

A reported $p = 0$ or $p < 0.001$ means that no sign-crossing bootstrap replicate was observed among the 10,000 draws; it is not a literal probability of zero. Holm–Bonferroni is applied separately to the six primary within-benchmark format contrasts (MATH-500, GSM8K, GPQA-Diamond). Each benchmark is a distinct inferential family because each corresponds to a separate task distribution; we do not pool the 18 contrasts for the primary analysis. A joint Holm correction over all 18 contrasts is reported only as a stricter sensitivity (Appendix Table 17). Two one-sided tests (TOST) at a ±1 pp equivalence margin use the 90% CI; this is an equivalence check, not a default claim. We use ±1 pp as a deliberately narrow practical sensitivity margin for this study: a one-percentage-point benchmark-accuracy difference is treated as operationally small for this comparison. The margin is a study-level tolerance, not a universal equivalence standard for reasoning benchmarks.

maj@5 McNemar (secondary). A problem is maj@5-correct if at least 3 of 5 MATH-500 seeds match gold. Exact McNemar tests this consensus estimand, not pass@1. Non-significant maj@5 tests do not imply that pass@1 effects are absent.

Pathology. A row is a *loop* if the longest consecutive identical-word run is ≥ 20 (Unicode word tokens, case-folded). A row is a *cap hit* if re-encoded completion length is $\geq$ 32,768. Because decode-then-re-tokenize undershoots the cap (observed max 32,737), we also count *near-cap* rows with $\geq$ 32,500 tokens. Saved rows do not include vLLM `finish_reason` or output token IDs; phrase-level cycles are invisible to the identical-word detector.

Length. Primary estimator: ratio of means over all seeds, plus the mean paired per-item token delta with a clustered percentile bootstrap CI. We also report the mean of per-item ratios, which is a different estimand and is dominated by short BF16 traces. Length is stratified by the 2×2 of BF16/quantized extractive correctness, with clustered 95% CIs on each cell. A direct clustered contrast $D = \overline{\Delta}_{\mathrm{BF16-only}} - \overline{\Delta}_{\mathrm{Both-OK}}$ uses the same problem-level resamples for both stratum means; replicates in which either stratum is empty are skipped. D compares conditional means and is not a causal effect. A BF16-internal control compares BF16 length when BF16 is correct versus incorrect. The BF16-correct conditional Δ, following Lian et al., is the mean quantized−BF16 Δ on (item, seed) pairs where BF16 is correct (Both-OK $\cup$ BF16-only). The Both-OK interval assesses whether jointly-correct traces show a detectable length change, whereas D assesses whether mismatch-associated lengthening exceeds jointly-correct lengthening. A D interval entirely above zero supports that conditional difference. The mismatch strata condition on which member of the pair is wrong. Because failure traces are themselves long, the large and oppositely signed BF16-only and quantized-only deltas are partly a consequence of this conditioning. We therefore use D as a diagnostic of correctness-conditioned mismatch asymmetry, not as an estimate of a causal quantization-induced token increase. We do not interpret a long BF16-failure tail as a Liu–Lian reconciliation.

Table 2: MATH-500 headline grid ($n = 500$, seeds 42–46). Clustered 95% CIs resample problems. Loops: identical-word run ≥ 20. Near-cap: completion tokens $\geq 32{,}500$. Exact cap hits after re-encoding: 0 in all cells.

Cell	**42**	**43**	**44**	**45**	**46**	**Mean ± Std**	**Clust. 95%**	**Tok.**	**L**	**NC**
Qwen BF16	94.4	94.0	93.8	94.6	93.2	94.00 ± 0.55	[92.2, 95.6]	4,011.4	0	14
Qwen FP8	94.4	95.2	94.8	92.6	95.0	94.40 ± 1.05	[92.8, 96.0]	4,007.7	0	15
Qwen AWQ-4	92.4	92.8	93.2	93.0	94.2	93.12 ± 0.67	[91.4, 94.8]	4,265.3	1	25
Qwen GPTQ-4	93.8	92.6	93.4	94.6	93.0	93.48 ± 0.77	[91.7, 95.2]	4,287.2	0	24
Llama BF16	89.0	88.4	90.2	89.8	88.8	89.24 ± 0.74	[87.2, 91.2]	4,656.6	3	19
Llama FP8	89.0	89.6	88.6	89.2	91.2	89.52 ± 1.01	[87.4, 91.6]	4,550.8	3	10
Llama AWQ-4	84.4	84.8	89.2	87.4	86.6	86.48 ± 1.96	[84.2, 88.7]	4,736.5	3	12
Llama GPTQ-4	88.0	89.6	86.8	89.4	90.8	88.92 ± 1.55	[86.9, 90.9]	4,840.5	2	21

Historical token-implied aggregate cost proxy. Following the Cost-of-Pass ratio form of Erol et al. [4], a historical token-implied aggregate cost proxy is

$$\text{Cost per query} = (\bar{T}/\tau) \times (R/3600), \tag{3}$$

divided by campaign pass@1, with $\tau = 65$ tok/s (unmeasured) and $R = \$1.50$ per A100-hour. Because τ and R are shared, ranking by this proxy is ranking by $\bar{T}$/pass@1. It is an aggregate sensitivity proxy, not a measured per-problem Cost-of-Pass quantity. We retain it only as a sensitivity (Appendix Table 16); primary systems/cost evidence is $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ in Section 3.4 and Table 11.

Observable modal agreement. Compact per-cell JSON stores `extractive_match` but not extracted strings. After recovering the 20,000 MATH-500 answer strings, we form agreement *without* gold. Campaign extraction and clustering used LightEval 0.8.0 with math-verify (frozen in `docs/ANSWER_NORMALIZATION.md`). A later throwaway MacBook install of LightEval 0.8.1 was *not* used for canonical numbers. For item i and five generated predictions, let $\sim$ be that campaign mathematical-equivalence relation. Cluster the five predictions into connected components of $\sim$. Let m_i be the size of the unique largest class when one exists; otherwise the item is a non-unique-mode tie. Observable agreement is $a_i = m_i/5$. Serve at threshold $k/5$ only when a unique modal class satisfies $m_i \geq k$. The pattern (A, A, B, B, C) has $m_i = 2$ and is not served at $\geq 3/5$; (A, A, A, B, B) has unique mode A with $m_i = 3$ and is served at $\geq 3/5$. Gold is used only afterward to estimate selective accuracy and risk. All five samples are generated before the decision. The five-sample token-cost proxy T_5 is the sum of their completion tokens, including abstained queries. We report three discrete operating points, not a continuous risk–coverage curve. Gold-hit ECE (confidence $= c_i/5$, label $= \mathbb{I}(c_i \geq 3)$) remains a deterministic function of the gold-hit histogram and is omitted.

4 Results

4.1 MATH-500 pass@1

Table 2 and Figure 5 show per-seed pass@1. FP8 point estimates sit 0.28–0.40 pp above BF16. The tested Llama AWQ-4 checkpoint shows the largest MATH-500 pass@1 drop among the evaluated Llama cells (-2.76 pp).

Table 3 is the test that matches RQ1. The tested Llama AWQ-4 checkpoint's clustered interval excludes zero ($p < 0.001$; Holm-significant among the six MATH contrasts). Qwen 4-bit MATH drops are negative but not Holm-significant. No MATH contrast, including FP8, has its 90% CI inside $[-1, +1]$ pp, so we do *not* claim statistical equivalence at a 1 pp serving

Table 3: Primary: problem-clustered bootstrap of MATH-500 pass@1 versus BF16 (B = 10,000). Secondary: exact McNemar on maj@5. Holm at $\alpha = 0.05$ over six contrasts (primary family). TOST ± 1 pp uses the 90% CI. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Contrast	Δ (pp)	**95% CI**	p	**Holm**	**90% CI**	**TOST**	**McNemar** p
Llama AWQ-4	-2.76	$[-4.16, -1.44]$	< 0.001	sig.	$[-3.92, -1.64]$	fail	0.296
Qwen AWQ-4	-0.88	$[-1.88, +0.08]$	0.069	n.s.	$[-1.68, -0.08]$	fail	1.000
Qwen GPTQ-4	-0.52	$[-1.40, +0.36]$	0.245	n.s.	$[-1.28, +0.20]$	fail	0.754
Qwen FP8	$+0.40$	$[-0.48, +1.28]$	0.383	n.s.	$[-0.32, +1.12]$	fail	0.581
Llama GPTQ-4	-0.32	$[-1.56, +0.96]$	0.619	n.s.	$[-1.36, +0.72]$	fail	1.000
Llama FP8	$+0.28$	$[-0.92, +1.48]$	0.647	n.s.	$[-0.72, +1.28]$	fail	1.000

Table 4: GSM8K clustered pass@1 versus matched BF16 (n = 1,319, 3 seeds). Holm over six contrasts. TOST ± 1 pp uses the 90% CI. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Contrast	Δ (pp)	**95% CI**	p	**Holm**	**90% CI**	**TOST**
Llama AWQ-4	-1.57	$[-2.55, -0.58]$	0.002	sig.	$[-2.38, -0.73]$	fail
Qwen AWQ-4	-0.20	$[-0.91, +0.48]$	0.579	n.s.	$[-0.78, +0.38]$	pass
Qwen GPTQ-4	-0.13	$[-0.86, +0.63]$	0.729	n.s.	$[-0.76, +0.51]$	pass
Llama GPTQ-4	$+0.28$	$[-0.53, +1.09]$	0.511	n.s.	$[-0.40, +0.96]$	pass
Qwen FP8	$+0.08$	$[-0.58, +0.73]$	0.827	n.s.	$[-0.45, +0.63]$	pass
Llama FP8	$+0.13$	$[-0.66, +0.94]$	0.761	n.s.	$[-0.53, +0.81]$	pass

margin. Secondary maj@5 McNemar p-values are all > 0.29: that consensus estimand does not capture the per-seed flips that pass@1 still sees.

4.2 GSM8K and GPQA-Diamond

Table 8 gives cell means; Tables 4 and 5 are the clustered tests. FP8 stays within about one point of BF16 on all three tasks; several GSM8K Qwen (and Llama FP8/GPTQ) contrasts pass TOST at ± 1 pp. The tested community Qwen AWQ-4 artifact showed 5.56 pp lower GPQA-Diamond pass@1 (95% CI $[-9.60, -1.52]$, $p = 0.007$; 3 seeds, $n = 198$). The Qwen AWQ GPQA result is significant within the primary Holm-6 family, but not under the Holm-18 joint sensitivity analysis (Appendix Table 17). Compact-JSON item counts show that drop is distributed across items rather than concentrated in a few problems: 75 of 198 GPQA items flip on at least one seed for the tested Qwen AWQ-4 artifact, and no item is BF16-correct / AWQ-wrong on all three seeds. The tested Llama AWQ-4 checkpoint also showed a GSM8K drop (-1.57 pp, 95% CI $[-2.55, -0.58]$, $p = 0.002$). GPQA intervals are otherwise wide; we treat non-AWQ GPQA contrasts as exploratory.

4.3 Loops and near-cap completions

Across 88 checkpoint$\times$benchmark$\times$seed runs the validator records **25** repetition-flagged rows (0.044% of 56,408) and **0** exact cap hits. Loops occur in BF16 as well as quantized cells (Llama MATH: 3/3/3/2 for BF16/FP8/AWQ/GPTQ). Three runs exceed 10,000 identical consecutive words. Near-cap completions ($\geq$ 32,500 tokens) total 209; on MATH-500, Qwen AWQ-4/GPTQ-4 have 25 and 24 versus BF16's 14. No row re-encoded to $\geq$ 32,768; native stop reasons were not logged.

Table 5: GPQA-Diamond clustered pass@1 versus matched BF16 ($n = 198$, 3 seeds). Holm over six contrasts. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant. The tested Qwen AWQ-4 row is the largest accuracy effect in this table; that Holm-6 result is not significant under the Holm-18 joint sensitivity (Appendix Table 17).

Contrast	Δ (pp)	**95% CI**	p	**Holm**	**90% CI**	**TOST**
Qwen AWQ-4	-5.56	$[-9.60, -1.52]$	0.007	sig.	$[-9.09, -2.19]$	fail
Qwen GPTQ-4	-2.36	$[-6.57, +1.85]$	0.288	n.s.	$[-6.06, +1.18]$	fail
Llama GPTQ-4	-1.18	$[-5.39, +3.03]$	0.573	n.s.	$[-4.71, +2.36]$	fail
Qwen FP8	-0.84	$[-4.71, +3.03]$	0.690	n.s.	$[-4.21, +2.53]$	fail
Llama AWQ-4	$+0.84$	$[-3.54, +5.05]$	0.711	n.s.	$[-2.86, +4.38]$	fail
Llama FP8	$+1.68$	$[-2.19, +5.39]$	0.403	n.s.	$[-1.52, +4.88]$	fail

Table 6: MATH-500 length versus BF16, all five seeds. RoM: ratio of means. Δ: mean paired per-item token difference with clustered 95% CI. 2×2 cells are mean quantized$-$BF16 Δ with clustered 95% CI and pair count n. BF16-correct column: BF16-correct conditional Δ, following Lian et al. BF16 len.: mean BF16 tokens when BF16 is correct vs. incorrect.

Contrast	**RoM**	**Mean Δ [95% CI]**	**Both OK**	**BF16 only**	**Quant only**	**Both wrong**	**BF16-correct Δ [95% CI]**
Qwen FP8	-0.1%	-4 $[-142, +134]$	$+51$ $[-42, +145]$ (n=2297)	$+5444$ $[+2922, +8035]$ (n=53)	-6940 $[-9479, -4345]$ (n=63)	$+264$ $[-1126, +1685]$ (n=87)	$+172$ $[+63, +291]$
Qwen AWQ-4	$+6.3\%$	$+254$ $[+116, +398]$	$+166$ $[+80, +254]$ (n=2272)	$+6915$ $[+4534, +9345]$ (n=78)	-6533 $[-8933, -4255]$ (n=56)	$+894$ $[-620, +2420]$ (n=94)	$+390$ $[+261, +531]$
Qwen GPTQ-4	$+6.9\%$	$+276$ $[+121, +431]$	$+136$ $[+45, +239]$ (n=2284)	$+6892$ $[+4090, +9896]$ (n=66)	-4910 $[-8180, -1715]$ (n=53)	$+1893$ $[+204, +3388]$ (n=97)	$+326$ $[+198, +464]$
Llama FP8	-2.3%	-106 $[-284, +65]$	$+38$ $[-74, +152]$ (n=2121)	$+2629$ $[+1098, +4265]$ (n=110)	-4159 $[-6040, -2428]$ (n=117)	-968 $[-2252, +198]$ (n=152)	$+166$ $[+33, +302]$
Llama AWQ-4	$+1.7\%$	$+80$ $[-109, +268]$	$+91$ $[-25, +210]$ (n=2052)	$+3090$ $[+1995, +4286]$ (n=179)	-4477 $[-6503, -2588]$ (n=110)	-305 $[-1520, +907]$ (n=159)	$+332$ $[+182, +490]$
Llama GPTQ-4	$+3.9\%$	$+184$ $[+12, +359]$	$+66$ $[-43, +181]$ (n=2097)	$+5915$ $[+4335, +7605]$ (n=134)	-4327 $[-6138, -2561]$ (n=126)	$+521$ $[-506, +1585]$ (n=143)	$+417$ $[+261, +583]$

4.4 Token length

On the *full* MATH-500 grid and all five seeds (Table 6, Figure 3), the tested Qwen AWQ-4/GPTQ-4 checkpoints showed +6.33%/+6.88% more tokens than BF16 (mean paired Δ +254 and +276 tokens; CIs exclude zero). Excluding near-cap pairs ($\geq$ 32,500 tokens) the ratios remain +4.61%/+5.75%. Llama GPTQ-4 is +3.95% (+184 tokens, CI just excludes zero). Llama AWQ-4 +1.72% and both FP8 cells have CIs that include zero.

A 200-item even-index, seed-42, mean-of-ratios analysis is a superseded estimator (Appendix G); it is not the full-grid ratio of means.

Figure 4 and Table 6 split paired token deltas by the 2×2. The primary descriptive length evidence is the full-grid ratio of means, the paired mean token Δ, and the Both-OK Δ among jointly correct pairs. Qwen AWQ-4 and GPTQ-4 Both-OK CIs exclude zero (+166 [+80, +254] and +136 [+45, +239]), so jointly-correct traces were still longer on those tested cells. The mismatch strata (BF16-only / Quant-only), mismatch-excess D, and the BF16 failure-length control are diagnostics. The large, oppositely signed mismatch-stratum deltas show that correctness-conditioned length summaries are strongly influenced by the long failure tail of whichever checkpoint is wrong. D therefore serves as a mismatch diagnostic (Table 7); it is not an estimate of a causal quantization-induced token increase. The Both-OK contrast gives the cleaner descriptive comparison among jointly correct outputs and is itself not causal. Qwen FP8 and all three Llama Both-OK CIs include zero. The BF16-correct conditional Δ, following Lian et al., is positive with CIs excluding zero in every contrast (+166 to +417 tokens). BF16-incorrect traces are themselves long: 15,843 vs. 3,256 tokens on Qwen and 13,928 vs. 3,539 on Llama. We report that control; we do not call it a Liu–Lian reconciliation.

Under the shared-τ proxy (Appendix Table 16), FP8 ranks first among Qwen cells. Measured Conditions A and B disagree with that order (Section 4.7).

4.5 Difficulty labels

Table 9: Level 1 is easy for every format in this sample. Among the evaluated Llama cells, the tested Llama AWQ-4 checkpoint showed the lowest pass@1 on Levels 2–4. MATH level labels

Table 7: Clustered mismatch excess $D = \overline{\Delta}_{\mathrm{BF16-only}} - \overline{\Delta}_{\mathrm{Both-OK}}$ on MATH-500. Both stratum means are formed on the same problem-level resamples (B = 10,000; empty-stratum replicates skipped; none occurred). D is a diagnostic of correctness-conditioned mismatch asymmetry, not a causal quantization-induced token increase.

Contrast	Both-OK Δ	BF16-only Δ	Mismatch excess D [95% CI]
Qwen FP8	$+51$ [$-42, +145$]	$+5444$ [$+2922, +8035$]	$+5394$ [$+2853, +7992$]
Qwen AWQ-4	$+166$ [$+80, +254$]	$+6915$ [$+4534, +9345$]	$+6748$ [$+4375, +9184$]
Qwen GPTQ-4	$+136$ [$+45, +239$]	$+6892$ [$+4090, +9896$]	$+6755$ [$+3938, +9786$]
Llama FP8	$+38$ [$-74, +152$]	$+2629$ [$+1098, +4265$]	$+2591$ [$+1047, +4244$]
Llama AWQ-4	$+91$ [$-25, +210$]	$+3090$ [$+1995, +4286$]	$+2999$ [$+1898, +4187$]
Llama GPTQ-4	$+66$ [$-43, +181$]	$+5915$ [$+4335, +7605$]	$+5849$ [$+4258, +7553$]

Table 8: Cross-benchmark pass@1 (mean ± seed std). MATH-500: 5 seeds. GSM8K and GPQA-Diamond: 3 seeds. L/NC: loops / near-cap over those seeds.

Cell	MATH-500	GSM8K	GPQA-D	L / NC
Qwen BF16	94.00 ± 0.55	91.26 ± 0.29	50.34 ± 2.96	4 / 28
Qwen FP8	94.40 ± 1.05	91.33 ± 0.16	49.49 ± 1.52	0 / 24
Qwen AWQ-4	93.12 ± 0.67	91.05 ± 1.14	44.78 ± 3.04	5 / 32
Qwen GPTQ-4	93.48 ± 0.77	91.13 ± 0.27	47.98 ± 1.75	3 / 40
Llama BF16	89.24 ± 0.74	88.68 ± 0.46	46.13 ± 1.91	3 / 21
Llama FP8	89.52 ± 1.01	88.80 ± 0.62	47.81 ± 0.29	3 / 12
Llama AWQ-4	86.48 ± 1.96	87.11 ± 0.23	46.97 ± 2.02	4 / 21
Llama GPTQ-4	88.92 ± 1.55	88.96 ± 0.70	44.95 ± 4.32	3 / 31

are not a monotone difficulty axis here (Level 2 Qwen BF16 89.11% vs Level 5 92.69%).

4.6 Observable modal agreement on MATH-500

Because MATH-500 was evaluated with five generations per item, we additionally examine whether observable answer agreement provides a useful gold-free abstention signal. Recovered answer strings make RQ3 measurable as observable multi-sample agreement, selective prediction, and abstention—not as model safety or softmax calibration, and not as G-Pass@k [18]. Table 10 reports coverage and selective risk at three discrete thresholds. On MATH-500, higher observable answer agreement is associated with lower selective error. At 5/5 consensus, observed selective error is at most 0.27% across the evaluated configurations, but coverage ranges from 70.2% (Llama AWQ-4) to 88.8% (Qwen FP8). Several 0.00% risk cells have bootstrap intervals $[0, 0]$; Wilson 95% upper bounds on those $0/n$ cells are 0.82%–1.08%. Zero observed errors is not zero true error.

The tested Llama AWQ-4 artifact showed lower strict-consensus coverage in some configurations: 70.2% versus 76.2% for BF16, a paired difference of -6.0 pp (95% CI $[-9.4, -2.6]$). Llama AWQ-4 $\geq 4/5$ coverage was 3.2 pp lower ($[-6.0, -0.6]$). Qwen AWQ-4 $\geq 4/5$ coverage was 2.2 pp lower ($[-4.0, -0.6]$). The $\geq 3/5$ AWQ contrasts, and all GPTQ-4 coverage contrasts, have paired intervals that include zero. No clear FP8–BF16 difference was detected in modal-agreement coverage or selective risk under these configurations. Appendix Table 12 lists the full threshold pattern.

The five-sample token-cost proxy is paid before abstention. Mean T_5 is 20,057 / 20,038 / 21,327 / 21,436 tokens for Qwen BF16/FP8/AWQ-4/GPTQ-4 and 23,283 / 22,754 / 23,682 / 24,203 for Llama. The tested Qwen 4-bit checkpoints retain an approximately +6–7% higher five-sample token sum in this accounting because T_5 is the sum of the same five traces. Tokens

Table 9: MATH-500 difficulty labels (5-seed pass@1 mean ± std). Labels are not monotone: Level 2 is harder than Level 5 for both BF16 backbones. Level-1 $n = 43$; treat cell-wise gaps as descriptive.

Level	**Qwen BF16**	**Qwen FP8**	**Qwen AWQ-4**	**Qwen GPTQ-4**
1 ($n = 43$)	98.60 ± 1.27	98.60 ± 2.08	98.14 ± 1.95	96.28 ± 2.08
2 ($n = 90$)	89.11 ± 0.93	90.44 ± 4.20	90.67 ± 1.69	88.67 ± 2.14
3 ($n = 105$)	95.81 ± 1.09	95.24 ± 1.51	94.10 ± 0.80	94.29 ± 1.90
4 ($n = 128$)	95.78 ± 0.89	96.41 ± 1.42	94.53 ± 1.24	96.56 ± 1.18
5 ($n = 134$)	92.69 ± 1.11	93.13 ± 1.23	91.04 ± 1.67	92.24 ± 1.95
Level	**Llama BF16**	**Llama FP8**	**Llama AWQ-4**	**Llama GPTQ-4**
1 ($n = 43$)	93.95 ± 3.12	94.88 ± 3.03	95.35 ± 1.64	96.28 ± 2.65
2 ($n = 90$)	84.67 ± 2.14	85.11 ± 2.02	81.11 ± 3.04	83.33 ± 3.42
3 ($n = 105$)	89.90 ± 2.48	90.10 ± 1.73	85.71 ± 1.17	89.71 ± 1.83
4 ($n = 128$)	91.56 ± 1.69	91.09 ± 0.43	87.03 ± 2.25	89.84 ± 1.75
5 ($n = 134$)	88.06 ± 1.75	88.81 ± 2.84	87.31 ± 3.46	88.81 ± 1.18

Table 10: Observable modal-answer agreement on MATH-500 ($n = 500$, seeds 42–46). Coverage is the fraction of items with a unique modal class of size $\geq k$. Risk is $1-$selective accuracy among served items. Mean T_5 is the five-sample token-cost proxy over all 500 items (abstained queries still pay). Gold is used only to score served predictions.

Model	**Fmt**	≥3/5 cov.	≥3/5 risk	≥4/5 cov.	≥4/5 risk	5/5 cov.	5/5 risk	Mean T_5
Qwen-7B	BF16	96.0	1.67	93.4	0.43	88.4	0.23	20,057
Qwen-7B	FP8	96.6	1.66	92.6	0.00	88.8	0.00	20,038
Qwen-7B	AWQ-4	95.8	1.46	91.2	0.44	86.4	0.23	21,327
Qwen-7B	GPTQ-4	95.4	1.47	92.8	0.43	86.8	0.00	21,436
Llama-8B	BF16	94.0	2.98	87.8	0.68	76.2	0.26	23,283
Llama-8B	FP8	94.0	3.19	88.6	0.90	78.2	0.00	22,754
Llama-8B	AWQ-4	93.2	3.65	84.6	1.89	70.2	0.00	23,682
Llama-8B	GPTQ-4	93.8	2.77	86.4	0.46	75.4	0.27	24,203

per served query divide that total by the number served, so abstained items are not free. This is not a measured monetary cost. Figure 1 shows the three operating points.

4.7 Measured serving throughput and Cost-of-Pass

RQ4 asks whether checkpoint rankings agree across the historical token proxy, sequential Condition A, and batched Condition B aggregate serving-cost proxies. They do not. Table 11 places Condition A and Condition B beside each other with $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ 95% intervals conditional on the observed timing repetitions and clustered accuracy bootstrap. Among Qwen cells the 65 tok/s proxy ranks FP8 < BF16 < AWQ-4 < GPTQ-4; Condition A ranks AWQ-4 < GPTQ-4 < FP8 < BF16; Condition B ranks GPTQ-4 < AWQ-4 < FP8 < BF16 (lower $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ first). Among Llama cells the proxy ranks FP8 first, Condition A ranks BF16 first, and Condition B ranks FP8 first. Different serving conditions produced different rankings; the A/B disagreement is serving-condition sensitivity, not an isolated batching effect. No unique lowest-cost checkpoint is identified across these estimators.

Qwen GPTQ-4 Condition B is -45.9% vs. Qwen BF16 (95% CI $[-46.4, -45.4]$) at the \$1.50/A100-hour pricing scenario. Qwen AWQ-4 Condition A has a lower aggregate hybrid cost proxy than Qwen GPTQ-4 (\$0.0215 vs. \$0.0242). Qwen FP8 Condition B mean $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ is -36.0% vs. BF16, but the five repeats show slow/mid/fast runtime variation at identical

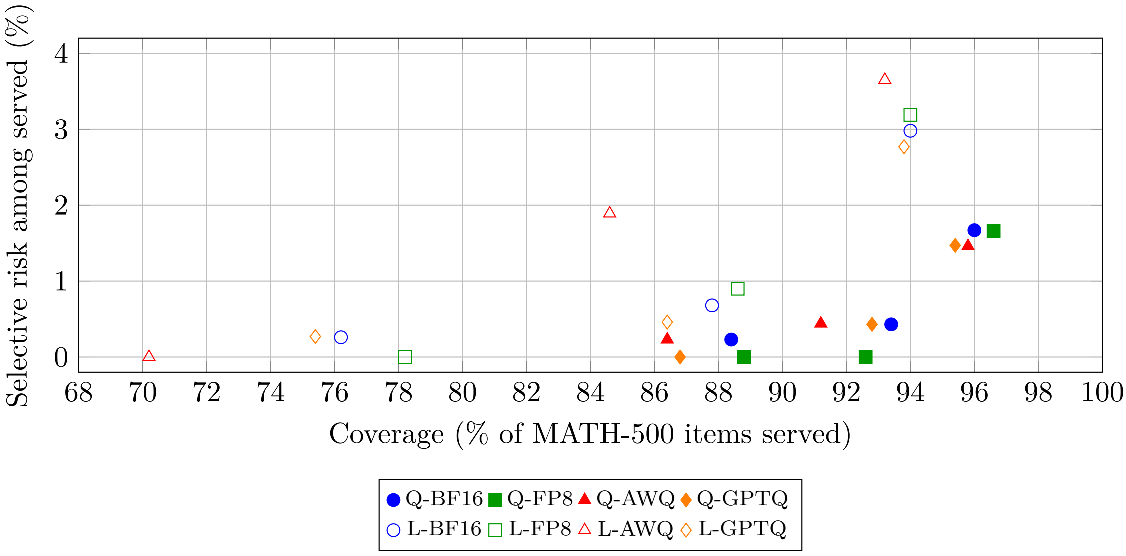

Figure 1: Three discrete MATH-500 operating points ($\geq 3/5$, $\geq 4/5$, $5/5$), not a continuous risk–coverage curve. Filled markers: Qwen-7B. Open markers: Llama-8B. Lower-right is high coverage and higher residual risk; lower-left is strict consensus.

aggregate output-token counts: two slow ~ 351 tok/s ($\widetilde{C}^{\text{hyb}}_{\text{pass}}$ \$0.0047), one mid ~ 456 tok/s (\$0.0036), two fast ~ 545 tok/s (\$0.0030). We keep all five; Appendix Table 15 lists them. The root cause of those timing regimes was not identified. Llama-8B FP8 Condition B is -8.2% $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ (95% CI $[-9.5, -6.9]$). Llama AWQ-4 and GPTQ-4 raise Condition B $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ (+18.5% and +24.9%). Llama GPTQ-4 Condition B mean throughput is about 0.2% lower than Llama BF16 (366.16 vs. 366.98 tok/s). These FP8 numbers are A100 Marlin W8A16, not Hopper W8A8.

Four-bit speed is not one number. Confirmation Condition A: Qwen AWQ-4 and GPTQ-4 exceed matched BF16 tok/s (76.89 and 82.67 vs. 43.92), whereas the tested Llama AWQ-4 and GPTQ-4 checkpoints do not (70.20 and 63.49 vs. 72.34). Batched Qwen GPTQ-4 reaches 488.26 ± 0.64 tok/s. Serving effects are checkpoint- and serving-condition-specific under this implementation.

A higher token count is not automatically cancelled by a higher measured tok/s. Cost is GPU-seconds $\times$ a pricing scenario, not tokens alone. No serving-confirmation *mean* output reached the 32,768-token generation cap. Compact confirmation JSON does not store native `finish_reason`.

Peak allocated VRAM is 53.9–56.1 GB for every batched confirmation cell. That is the 0.75 utilization pool, not a measurement of 4-bit weight-footprint savings. We do not publish a format-to-format weight table from `torch.cuda.memory_allocated` after engine init.

Figure 2 is a Condition B cost-accuracy scatter. Panels represent separate serving conditions (different hosts and model families) and are not a pooled trade-off plot. Latency detail for Condition A remains in Appendix Table 13.

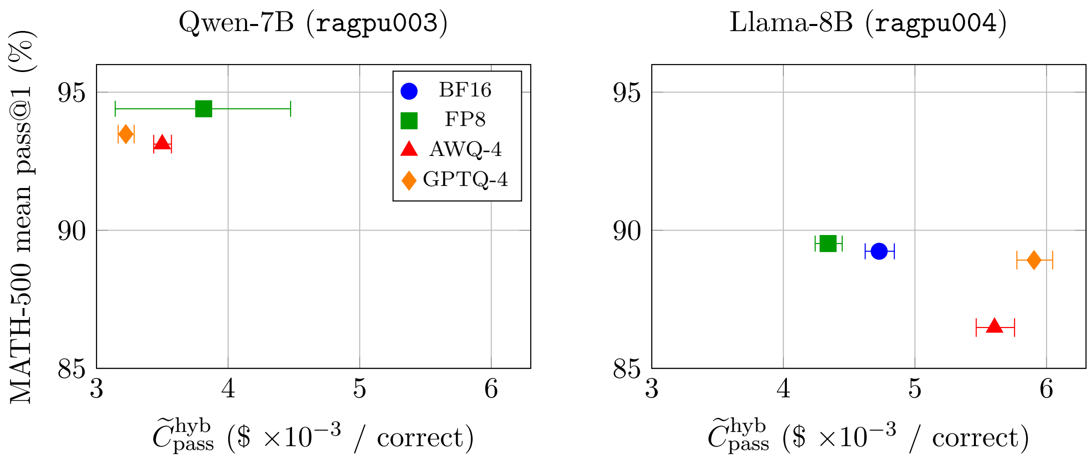

Figure 2: Condition B cost-accuracy scatter: aggregate hybrid Cost-of-Pass proxy versus campaign MATH-500 pass@1 (\$ $\times 10^{-3}$). Whiskers are 95% uncertainty intervals conditional on the observed timing repetitions and clustered accuracy bootstrap. Points are not joined. Panels represent separate serving conditions (Qwen-7B on `ragpu003`; Llama-8B on `ragpu004`) and are not a pooled trade-off plot. Qwen FP8's wide whisker is the five-rep wall-clock spread.

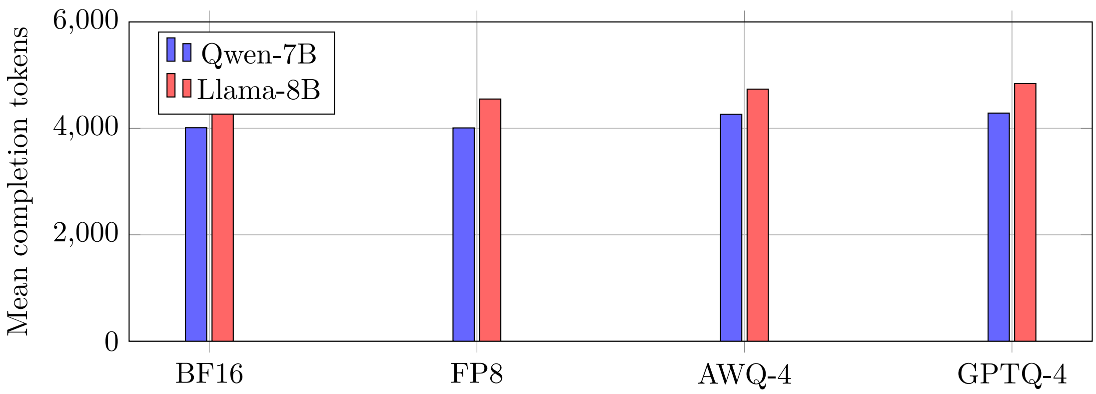

Figure 3: Full-grid mean MATH-500 completion tokens (ratio-of-means estimand). Bars are point estimates; clustered 95% intervals on paired token Δ are in Table 6 and are not drawn on this axis.

Table 11: Aggregate hybrid Cost-of-Pass proxy $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ on the controlled confirmation. Condition A: 20 sequential prompts. Condition B: 100-prompt `llm.generate`, `max_num_seqs=8`. Tok/s is mean± sample SD. The numerator is confirmation GPU-sec/query; the denominator is campaign MATH-500 pass@1 at \$1.50/A100-hour. Intervals are 95% (timing-rep $\times$ clustered pass@1), conditional on the observed timing repetitions. Qwen: `ragpu003`. Llama: `ragpu004`.

Model	**Fmt**	**A tok/s**	**A GPU-s/q**	**A $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ [95% CI]**	**B tok/s**	**B GPU-s/q**	**B $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ [95% CI]**	**B Δ% vs BF16**
Qwen-7B	BF16	43.92 ± 0.04	102.51	\$0.0454 [0.0447, 0.0463]	252.72 ± 0.56	13.44	\$0.0060 [0.0059, 0.0061]	anchor
Qwen-7B	FP8	62.50 ± 0.08	84.60	\$0.0373 [0.0367, 0.0380]	449.79 ± 97.53	8.64	\$0.0038 [0.0031, 0.0045]	−36.0 [−47.2, −24.8]
Qwen-7B	AWQ-4	76.89 ± 0.70	48.04	\$0.0215 [0.0211, 0.0219]	418.15 ± 2.03	7.82	\$0.0035 [0.0034, 0.0036]	−41.3 [−41.9, −40.6]
Qwen-7B	GPTQ-4	82.67 ± 0.69	54.39	\$0.0242 [0.0238, 0.0248]	488.26 ± 0.64	7.23	\$0.0032 [0.0032, 0.0033]	−45.9 [−46.4, −45.4]
Llama-8B	BF16	72.34 ± 0.27	66.43	\$0.0310 [0.0303, 0.0318]	366.98 ± 1.90	10.13	\$0.0047 [0.0046, 0.0048]	anchor
Llama-8B	FP8	78.75 ± 2.89	92.55	\$0.0431 [0.0415, 0.0450]	481.42 ± 1.36	9.32	\$0.0043 [0.0042, 0.0044]	−8.2 [−9.5, −6.9]
Llama-8B	AWQ-4	70.20 ± 2.71	89.51	\$0.0431 [0.0415, 0.0450]	391.11 ± 0.48	11.63	\$0.0056 [0.0055, 0.0058]	+18.5 [+16.6, +20.5]
Llama-8B	GPTQ-4	63.49 ± 0.41	86.95	\$0.0407 [0.0398, 0.0418]	366.16 ± 0.79	12.60	\$0.0059 [0.0058, 0.0060]	+24.9 [+23.0, +26.7]

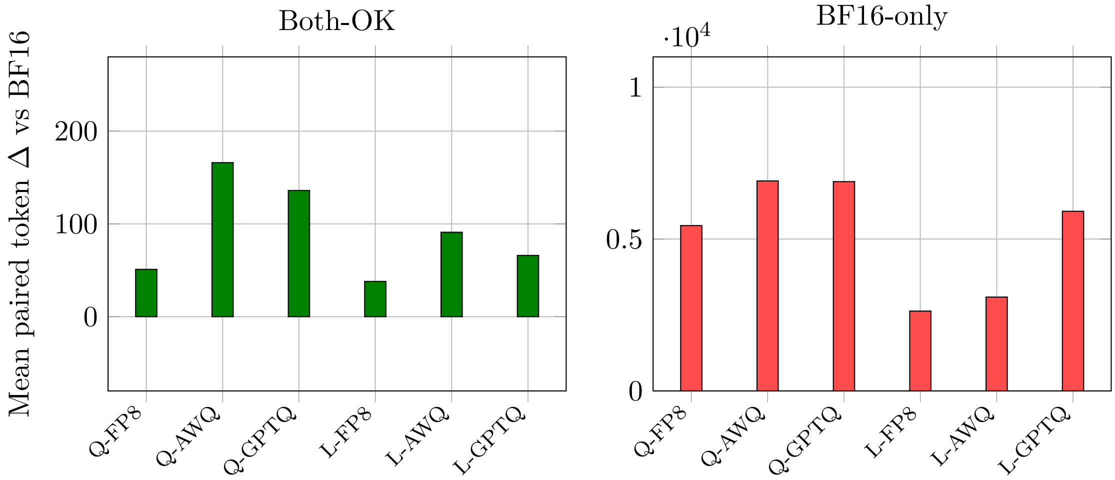

Figure 4: Paired token deltas by correctness, two panels so Both-OK is visible. Left: jointly-correct pairs (Qwen AWQ/GPTQ CIs exclude 0; FP8 and Llama CIs include 0; n in Table 6). Right: BF16-correct / quantized-wrong pairs. Clustered 95% CIs and n are in Table 6.

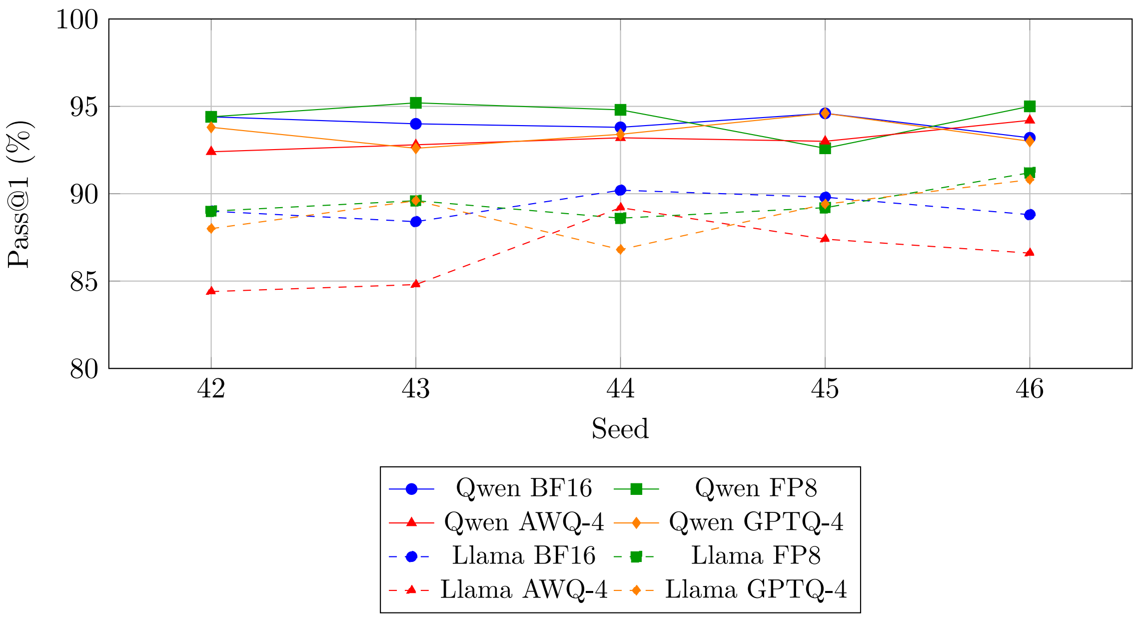

Figure 5: MATH-500 pass@1 by seed. Solid: Qwen-7B. Dashed: Llama-8B. The tested Llama AWQ-4 checkpoint shows the largest MATH-500 pass@1 drop among the evaluated Llama cells.

5 Discussion

Under this pinned A100 W8A16 stack, evaluating these eight public checkpoints requires more than a single pass@1 or token count. FP8–BF16 MATH intervals include zero, but TOST at ± 1 pp fails, so we do not treat FP8 as a statistically equivalent accuracy default. Four-bit behavior is checkpoint- and task-specific: the tested Qwen AWQ-4 checkpoint keeps MATH-500 and GSM8K near BF16 while showing 5.56 pp lower pass@1 on GPQA-Diamond. The Qwen AWQ GPQA result is significant within the primary Holm-6 family, but not under the Holm-18 joint sensitivity analysis. The tested Llama AWQ-4 checkpoint shows the largest MATH-500 and GSM8K pass@1 drops among the evaluated Llama cells. The tested Qwen 4-bit checkpoints showed 6.3–6.9% higher mean MATH-500 length. Both-OK CIs exclude zero for those Qwen 4-bit cells, so jointly-correct traces were still longer. Mismatch-conditioned differences are much larger because failure traces are themselves long; D is a diagnostic of that asymmetry, not a causal compression effect.

Pass@1 and maj@5 are different estimands. The tested Llama AWQ-4 artifact's lower pass@1 appears in both mean accuracy and 5/5 consensus coverage. The tested Qwen 4-bit checkpoints retain relatively strong MATH-500 pass@1 while emitting more tokens. No clear FP8–BF16 difference was detected in modal-agreement coverage or selective risk under these configurations; paired intervals that include zero do not establish equivalence. Repeated sampling is associated with lower selective risk on MATH-500 but costs about five generations per attempted item. Wilson intervals on $0/n$ cells remain strictly positive. The construction is a gold-free unique-mode abstention rule on five MATH seeds, not G-Pass@k and not a safety property.

A 200-item even-index length estimator, an unconstrained serving timing, and a vLLM 0.8.5 pathology autopsy are superseded analyses (Appendix G); they are not mixed with the canonical 56k grid or the confirmation serving numbers.

Cost estimators disagree. The historical 65 tok/s token proxy, sequential Condition A, and batched Condition B rank the four Qwen (resp. Llama) cells differently. Qwen GPTQ-4 Condition B $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ is -45.9% vs. BF16, but Condition A ranks Qwen AWQ-4 first, and Qwen FP8 Condition B is a five-rep mixture of slow and fast wall-clock regimes whose cause was not identified. That A/B disagreement is serving-condition sensitivity (workload and serving regime jointly), not an isolated batching effect. Llama GPTQ-4 Condition B mean throughput is about 0.2% lower than Llama BF16. Peak allocated VRAM did not separate formats under the 0.75 utilization cap and reflects the engine allocation pool rather than isolated weight-footprint compression.

The deployment implication is not that one quantization format is universally preferable. Public checkpoints should be evaluated on the target task and intended serving condition: token-based proxies can mis-rank measured GPU-seconds, different serving conditions produced different rankings, and the tested community quantization artifacts can show task-specific accuracy losses. Deployment decisions should therefore combine checkpoint-specific accuracy with direct measurements from the intended serving configuration. The contribution is a stack-pinned measurement protocol showing that public quantized reasoning checkpoints can change rank depending on what is measured.

6 Limitations

1. **A100 W8A16 fallback.** FP8 checkpoints were executed as Marlin W8A16 on A100, not native W8A8. Numbers do not transfer to Hopper native W8A8.

2. **Measured serving scope.** Throughput, latency, GPU-sec/query, and $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ are measured on balanced MATH-500 serving subsets (20 sequential prompts for Condition A; 100 batched prompts for Condition B), not the full 56k campaign. Pass@1 in the proxy

denominator is the 40-cell campaign mean. Both serving conditions use difficulty-balanced subsets rather than the natural MATH-500 level distribution. Conditions A and B also differ jointly in prompt subset and serving regime. Because continuous-batch Condition B records aggregate elapsed time rather than per-request or per-level service time, the retained data cannot be retrospectively reweighted to identify natural-distribution GPU-seconds, and the A/B ranking reversal should be interpreted as serving-condition sensitivity rather than a causal batching effect. The \$1.50/A100-hour rate is a pricing scenario, not billed cluster cost. Confirmation Condition B pins `max_num_seqs=8`. Repetitions share sampling seed 20260816. Qwen and Llama confirmation jobs used different hosts (`ragpu003` vs. `ragpu004`); within-architecture comparisons are same-node. Qwen FP8 Condition B retains five wall-clock repeats with large timing variance at identical aggregate output-token counts; the cause was not identified. Peak allocated VRAM is the 0.75 engine pool, not isolated weight footprint. Compact confirmation JSON has no `finish_reason`.

3. **Pathology metrology.** The loop detector is identical-word runs ≥ 20. Phrase-level cycles and `finish_reason` were not saved in the original campaign. Near-cap is a proxy.

4. **Modal agreement.** Five generations are required per attempted MATH problem. Only MATH-500 has five seeds in this campaign. Mathematical answer equivalence is evaluator- and task-specific (campaign LightEval 0.8.0 / math-verify). High agreement can still be wrong. The signal is not general uncertainty calibration, not softmax calibration, and not a safety property.

5. **Checkpoint provenance.** AWQ-4 is a community quantization of unknown calibration set and quantizer revision. Accuracy losses on the tested community AWQ artifacts are properties of those artifacts, not a proof about AWQ as a method. GPTQ-4/FP8 are RedHatAI artifacts (appendix).

6. **Scope.** Two 7–8B DeepSeek-R1 distillations, English math/science, one pinned serving stack, one prompt. GSM8K/GPQA use three seeds.

7. **Not evaluated.** This study does not evaluate AIME, LiveCodeBench, 32B models, Hopper native W8A8 FP8, energy measurements, production continuous batching, or probability calibration.

8. **Reproducibility surface.** Compact per-cell JSON, recovered MATH-500 answer strings, confirmation timing JSON, and analysis reports are released. Full GPU traces (chain-of-thought text) are not publicly released. Compact campaign JSON also omits `finish_reason` and output token IDs. Tables are reproducible from the released compact artifacts via the documented `--check` commands. Replaying the complete GPU campaign requires the original A100 hardware, checkpoint weights, and dataset revisions; the protocol is inspectable and is not expected to be rerun by every reviewer.

9. **Contamination.** MATH and GSM8K overlap public pretraining corpora. A paired within-item design reduces but does not eliminate contamination interaction with checkpoint.

10. **Source configs vs. launcher.** Files in `configs/legacy_models/` are historical harness configs and are not the QRM `inference.py` campaign. Effective KV-cache dtype is not stored per cell. Launcher flags such as `max_model_len=32768` are code/launcher-confirmed, not per-job runtime telemetry.

Threats to validity. Construct: pass@1, maj@5, token length, GPU-seconds, and the aggregate hybrid Cost-of-Pass proxy are distinct estimands; the hybrid proxy is not Erol et al.'s per-problem Cost-of-Pass, and identical-word loops are not `finish_reason`. Internal: Conditions A and B are complete serving-condition contrasts (subset and regime jointly), not a pure batching ablation; Qwen and Llama confirmation jobs used different hosts; the \$1.50/A100-hour figure is a pricing scenario. External: two 7–8B public R1-distill families, one A100 / vLLM 0.7.0 eager stack, and three English math/science benchmarks; results do not transfer to Hopper native W8A8, 32B models, AIME, LiveCodeBench, energy, or production continuous batching.

7 Conclusion

Under a pinned A100 / vLLM 0.7.0 stack, these eight public R1-distill checkpoints are not ranked by a single pass@1 or token count. Clustered pass@1, maj@5, Both-OK versus BF16-correct conditional length, and the historical token proxy versus sequential versus batched $\widetilde{C}_{\text{pass}}^{\text{hyb}}$ can disagree. The tested Llama AWQ-4 checkpoint shows the largest MATH-500 and GSM8K pass@1 drops among the evaluated Llama cells; the tested community Qwen AWQ-4 artifact showed the largest GPQA-Diamond drop among the evaluated Qwen cells. The Qwen AWQ GPQA result is significant within the primary Holm-6 family, but not under the Holm-18 joint sensitivity analysis. FP8–BF16 MATH intervals include zero but fail ± 1 pp TOST. The tested Qwen 4-bit checkpoints showed 6.3–6.9% higher MATH-500 mean length, with Both-OK CIs excluding zero; mismatch-conditioned differences are much larger because failure traces are themselves long. Identical-word repetition flags are rare under our detector (25/56,408); near-cap completions are more common for Qwen 4-bit (25/24 vs. 14). Gold-free 5/5 unique-mode abstention has low observed selective risk and strictly positive Wilson upper bounds on $0/n$ cells. Cost-estimator rankings disagree across the 65 tok/s proxy, Condition A, and Condition B; Qwen FP8 Condition B is a five-rep wall-clock mixture, not a lone -36.0%. We provide a stack-pinned measurement protocol showing that public quantized reasoning checkpoints can change rank depending on what is measured. The deployment ranking of a quantized reasoning checkpoint is not a property of bit-width alone.

Artifacts

Code, patches, validation JSON, and analysis reports: Manish06N/reasoning-compression-lab. Per-cell records: `results/math500/`, `results/gsm8k/`, `results/gpqa/`. Canonical pass@1 tables: `revision_reanalysis_report.json`. Modal agreement: `modal_agreement_report.json`. Measured serving (confirmation): `results/reports/measured_serving_confirmation/`. First unconstrained timing (provenance only): `results/measured_serving/`. See `paper/ARTIFACT.md`. GPQA is a gated benchmark; public records must not republish item text.

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CRediT authorship contribution statement

Conceptualization: Manish Nandish, Rajiv Misra. **Methodology:** Manish Nandish. **Software:** Manish Nandish. **Investigation:** Manish Nandish. **Data curation:** Manish Nandish.

Formal analysis: Manish Nandish. **Visualization:** Manish Nandish. **Writing – original draft:** Manish Nandish. **Writing – review & editing:** Manish Nandish, Rajiv Misra, Midhunchakkaravarthy Janarthanan. **Supervision:** Rajiv Misra, Midhunchakkaravarthy Janarthanan. **Resources:** Rajiv Misra.

A Prompt

MATH-500 and the GSM8K/GPQA QRM templates use:

```
{question}

Please reason step by step, and put your final
answer within \boxed{}.
```

with a <think> prefix on the official QRM path.

B Checkpoints

Cell	Hugging Face ID / revision
Qwen BF16	`deepseek-ai/DeepSeek-R1-Distill-Qwen-7B` `916b56a44061fd5cd7d6a8fb632557ed4f724f60`
Qwen FP8	`RedHatAI/DeepSeek-R1-Distill-Qwen-7B-FP8-dynamic` `ceb2fcd178d477cd21b92ffb43164c74165a212c`
Qwen AWQ-4	`jakiAJK/DeepSeek-R1-Distill-Qwen-7B_AWQ` `1d082b3ded4e7cefbb6b32cd5e07c7d409832816`
Qwen GPTQ-4	`RedHatAI/DeepSeek-R1-Distill-Qwen-7B-quantized.w4a16` `9559a91ed463eecda64b93a532fe5aa8cc25f2eb`
Llama BF16	`deepseek-ai/DeepSeek-R1-Distill-Llama-8B` `6a6f4aa4197940add57724a7707d069478df56b1`
Llama FP8	`RedHatAI/DeepSeek-R1-Distill-Llama-8B-FP8-dynamic` `5d548d9169c53b7bb7ef0b3bc261509d5da6e3dd`
Llama AWQ-4	`jakiAJK/DeepSeek-R1-Distill-Llama-8B_AWQ` `57360e84a0e7278c6b3abb453f2dc812b213badb`
Llama GPTQ-4	`RedHatAI/DeepSeek-R1-Distill-Llama-8B-quantized.w4a16` `faa06ed0a4a5bdab76c4df12376e0c5228c0bdfe`

AWQ-4 checkpoints are third-party community uploads; calibration corpus and quantization code revision are unknown. Accuracy losses on the tested community AWQ artifacts are properties of those artifacts, not a proof about AWQ as a method. GPTQ-4 and FP8 are RedHatAI compressed-tensors artifacts released for vLLM; this study evaluates those public checkpoints under A100 Marlin W8A16 and does not report Red Hat production serving measurements. Dataset revision SHAs for MATH-500 / GSM8K / GPQA-Diamond are recorded in `results/reports/runtime_manifest.json` and `configs/tasks/`.

Serving confirmation protocol and raw JSON live under `docs/MEASURED_SERVING_CONFIRMATION_PROTOCOL.md` and `results/measured_serving_confirmation/raw/`. Aggregates: `measured_serving_confirmation_report.json`. An earlier unconstrained timing (`docs/MEASURED_SERVING_PROTOCOL.md`, `results/measured_serving/`) is provenance only and is not mixed with confirmation means.

C Reanalysis

Pathology, clustered bootstrap, and token strata are produced by the canonical reanalysis script from the released per-cell JSON. Loop threshold = 20. Near-cap threshold = 32,500. Bootstrap $B = 10{,}000$, seed 0.

Table 12: Paired coverage Δ versus matched BF16 (percentage points). Intervals that exclude zero are marked *. Selective-accuracy Δ intervals are in the machine-readable modal report.

Contrast	$\geq$3/5 cov. Δ [95% CI]	$\geq$4/5 cov. Δ [95% CI]	5/5 cov. Δ [95% CI]
Qwen FP8	+0.6 [−0.8, +2.0]	−0.8 [−2.2, +0.6]	+0.4 [−1.6, +2.4]
Qwen AWQ-4	−0.2 [−1.8, +1.2]	−2.2 [−4.0, −0.6]*	−2.0 [−4.4, +0.2]
Qwen GPTQ-4	−0.6 [−2.0, +0.8]	−0.6 [−2.2, +1.0]	−1.6 [−4.2, +0.8]
Llama FP8	−0.0 [−1.8, +1.8]	+0.8 [−1.6, +3.2]	+2.0 [−1.2, +5.2]
Llama AWQ-4	−0.8 [−2.8, +1.2]	−3.2 [−6.0, −0.6]*	−6.0 [−9.4, −2.6]*
Llama GPTQ-4	−0.2 [−2.2, +1.8]	−1.4 [−3.8, +1.0]	−0.8 [−4.2, +2.6]

Table 13: Confirmation single-stream serving (Condition A: 20 sequential prompts). Latency columns are means of per-run median/P90. $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ uses campaign MATH-500 pass@1 and a \$1.50/A100-hour scenario. Tok/s is mean± sample SD.

Model	**Fmt**	**tok/s**	**Med. lat (s)**	**P90 lat (s)**	**GPU-s/q**	$\widetilde{C}^{\text{hyb}}_{\text{pass}}$
Qwen-7B	BF16	43.92 ± 0.04	59.19	244.83	102.51	\$0.0454
Qwen-7B	FP8	62.50 ± 0.08	43.60	197.01	84.60	\$0.0373
Qwen-7B	AWQ-4	76.89 ± 0.70	26.47	110.43	48.04	\$0.0215
Qwen-7B	GPTQ-4	82.67 ± 0.69	34.78	67.51	54.39	\$0.0242
Llama-8B	BF16	72.34 ± 0.27	32.40	126.10	66.43	\$0.0310
Llama-8B	FP8	78.75 ± 2.89	38.00	253.55	92.55	\$0.0431
Llama-8B	AWQ-4	70.20 ± 2.71	40.12	246.14	89.51	\$0.0431
Llama-8B	GPTQ-4	63.49 ± 0.41	46.31	242.83	86.95	\$0.0407

D Modal-agreement details

The compact recovered artifact has 20,000 rows (SHA256 prefix `23e9ead021111959`; full digest is in the JSON sidecar). It stores extracted prediction representations, campaign `extractive_match`, and completion tokens. It does not store chain-of-thought or problem text. Campaign extraction used LightEval 0.8.0; a later MacBook LightEval 0.8.1 install was not used for these numbers. Campaign extractive match was reproduced on 20,000/20,000 rows before clustering. Equivalence clustering reported 0 symmetry violations, 0 transitivity violations, and 116 non-unique-mode ties among 4,000 groups. Bootstrap uses $B = 10{,}000$ problem resamples with seed 20260816. Secondary ECE/Brier scores (confidence $= m_i/5$) are in the JSON report and are not treated as a primary calibration result.

T_5 is the five-sample token-cost proxy: the sum of completion tokens over seeds 42–46. Mean / median / P90 T_5 are computed over all 500 items. Tokens per served query and tokens per correct served result use that same 500-item T_5 total in the numerator, so abstained queries are not free.

E Measured serving: latency, microbenchmark, FP8 repeats, and token proxy

Condition A latency detail (Table 13) is sequential decode of a balanced 20-prompt MATH-500 subset. Median latency is the mean of per-run medians. $R = 3$ except Llama AWQ-4 ($R = 5$; CV-triggered). Fixed-token microbenchmark (Table 14): 10 prompts $\times$ 512 tokens, $T = 0.0$, `ignore_eos`. Table 15 lists all five Qwen FP8 Condition B repeats. Table 16 is the historical 65 tok/s token proxy, not measured wall-clock.

Table 14: Fixed-token decode microbenchmark (10 prompts × 512 tokens). Isolates kernel decode speed from trace-length variation. Secondary to task-realistic Conditions A/B.

Model	Fmt	Raw decode tok/s
Qwen-7B	BF16	205.41
Qwen-7B	FP8	437.19
Qwen-7B	AWQ-4	377.73
Qwen-7B	GPTQ-4	401.19
Llama-8B	BF16	362.01
Llama-8B	FP8	392.05
Llama-8B	AWQ-4	344.04
Llama-8B	GPTQ-4	302.54

Table 15: Qwen-7B FP8 Condition B: all five technical repeats (identical aggregate 373,794 output tokens). Slow: tok/s < 400. Fast: tok/s > 500. $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ uses campaign pass@1 0.944. The cause of the timing regimes was not identified.

Rep	tok/s	GPU-s/q	$\widetilde{C}^{\text{hyb}}_{\text{pass}}$	Regime
1	350.91	10.6522	\$0.0047	slow
2	350.68	10.6591	\$0.0047	slow
3	455.90	8.1990	\$0.0036	mid
4	545.18	6.8563	\$0.0030	fast
5	546.26	6.8427	\$0.0030	fast
Mean ± SD	449.79 ± 97.53	—	\$0.0038	keep all five
Median / IQR	455.90 / [350.91, 545.18]	—	—	—

F Holm-18 sensitivity

Primary multiplicity control is Holm–Bonferroni within each benchmark's six format contrasts. Table 17 applies Holm jointly to all 18 contrasts as a stricter sensitivity. Llama MATH AWQ-4 and Llama GSM8K AWQ-4 remain significant. The tested community Qwen AWQ-4 artifact showed 5.56 pp lower GPQA-Diamond pass@1. The Qwen AWQ GPQA result is significant within the primary Holm-6 family, but not under the Holm-18 joint sensitivity analysis (Holm-adjusted $p = 0.109$). That does not change the primary within-benchmark inference.

G Superseded analyses

The following notes are retained only so old links do not 404; they are not mixed with canonical numbers.

200-item even-index length. A previous "stratified mixed-correctness" analysis used even indices among the first 400 MATH-500 problems, seed 42 only, and the mean of per-item ratios. On those 200 items it reports Qwen FP8 +11.4% while the ratio of means is −3.4%. The canonical estimator is the full-grid ratio of means plus clustered paired Δ (Table 6).

Unconstrained serving timing. `results/measured_serving/` used unpinned `max_num_seqs`, an unbalanced Condition A first-20 draw, and mixed-node Llama repeats. Absolute tok/s from that run is not comparable to the confirmation after pinning concurrency 8. Those files are provenance; they are not averaged with confirmation and are not cited in the abstract.

Table 16: Historical token-implied aggregate cost proxy on MATH-500 at \$1.50/A100-hour and a shared 65 tok/s. Not measured wall-clock. Ranking equals ranking by mean tokens / pass@1. Primary cost evidence is Table 11.

Cell	Mean tok.	Cost / Q	Token proxy
Qwen BF16	4,011.4	\$0.02571	\$0.02736
Qwen FP8	4,007.7	\$0.02569	\$0.02721
Qwen AWQ-4	4,265.3	\$0.02734	\$0.02936
Qwen GPTQ-4	4,287.2	\$0.02748	\$0.02940
Llama BF16	4,656.6	\$0.02985	\$0.03345
Llama FP8	4,550.8	\$0.02917	\$0.03259
Llama AWQ-4	4,736.5	\$0.03036	\$0.03511
Llama GPTQ-4	4,840.5	\$0.03103	\$0.03490

Table 17: Secondary Holm correction over all 18 pass@1 contrasts. Primary inference remains Holm-6 within each benchmark. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Bench.	Contrast	p	Holm-6	Holm-18 p	Holm-18
MATH	Llama AWQ-4	< 0.001	sig.	< 0.001	sig.
MATH	Qwen AWQ-4	0.069	n.s.	1.000	n.s.
MATH	Qwen GPTQ-4	0.245	n.s.	1.000	n.s.
MATH	Qwen FP8	0.383	n.s.	1.000	n.s.
MATH	Llama GPTQ-4	0.619	n.s.	1.000	n.s.
MATH	Llama FP8	0.647	n.s.	1.000	n.s.
GSM8K	Llama AWQ-4	0.002	sig.	0.031	sig.
GSM8K	Llama GPTQ-4	0.511	n.s.	1.000	n.s.
GSM8K	Qwen AWQ-4	0.579	n.s.	1.000	n.s.
GSM8K	Qwen GPTQ-4	0.729	n.s.	1.000	n.s.
GSM8K	Llama FP8	0.761	n.s.	1.000	n.s.
GSM8K	Qwen FP8	0.827	n.s.	1.000	n.s.
GPQA	Qwen AWQ-4	0.007	sig.	0.109	n.s.
GPQA	Qwen GPTQ-4	0.288	n.s.	1.000	n.s.
GPQA	Llama FP8	0.403	n.s.	1.000	n.s.
GPQA	Llama GPTQ-4	0.573	n.s.	1.000	n.s.
GPQA	Qwen FP8	0.690	n.s.	1.000	n.s.
GPQA	Llama AWQ-4	0.711	n.s.	1.000	n.s.

Pathology key mismatch. Earlier drafts reported 0/0 loops and cap hits because analysis scripts read `truncation_count` / `repetition_flag_count` while the validator writes `token_limit_hits` / `repetition_rows`. Canonical counts are 25 loops and 0 exact cap hits over 56,408 completions (Section 4.3).

vLLM 0.8.5 pathology autopsy. Earlier drafts wrote that pinning vLLM 0.7.0 "removed" truncations and loops seen on vLLM 0.8.5. Those earlier runs also differed in harness, decoding flags (including a bug that dropped `repetition_penalty`), and prompt profile. The final grid still contains 25 loops. We do not attribute pathology rates causally to the engine version. A clean stack × format factorial remains future work.

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